

The slide features decorative blue abstract shapes in the corners. The top-left corner has a solid blue shape with a thick blue curved line. The top-right corner has a light blue wavy line. The bottom-left corner has a light blue wavy line. The bottom-right corner has a solid blue shape with a thick blue curved line.

Midterm Mastery Presentation

Present By Yalin Sun

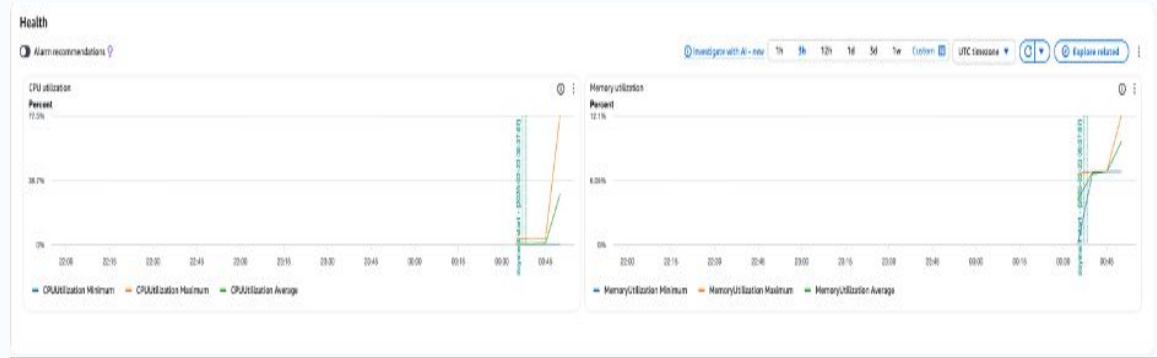
Email: sun.yalin@northeastern.edu

Step I: HW 6 Results

Part II: Identifying Performance Bottlenecks!

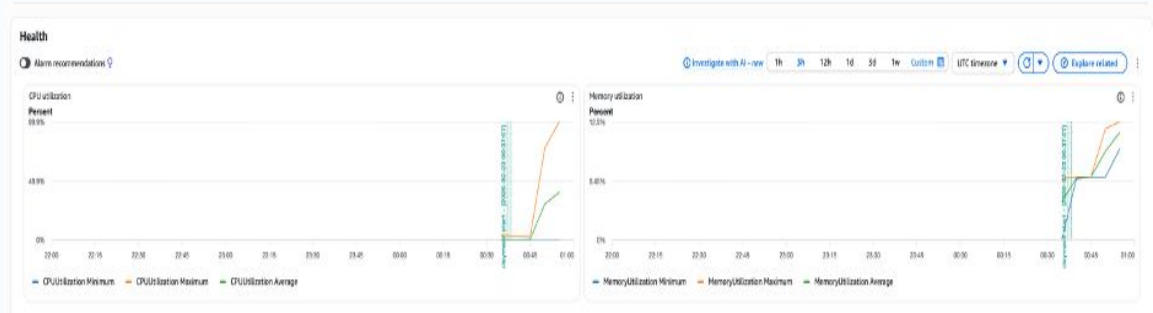
Test 1: Baseline Load

- Users: 5
- Spawn Rate: 5 users/second
- Run Time: 2 minutes



Test 2: Breaking Point

- Users: 20
- Spawn Rate: 20 users/second
- Run Time: 3 minutes

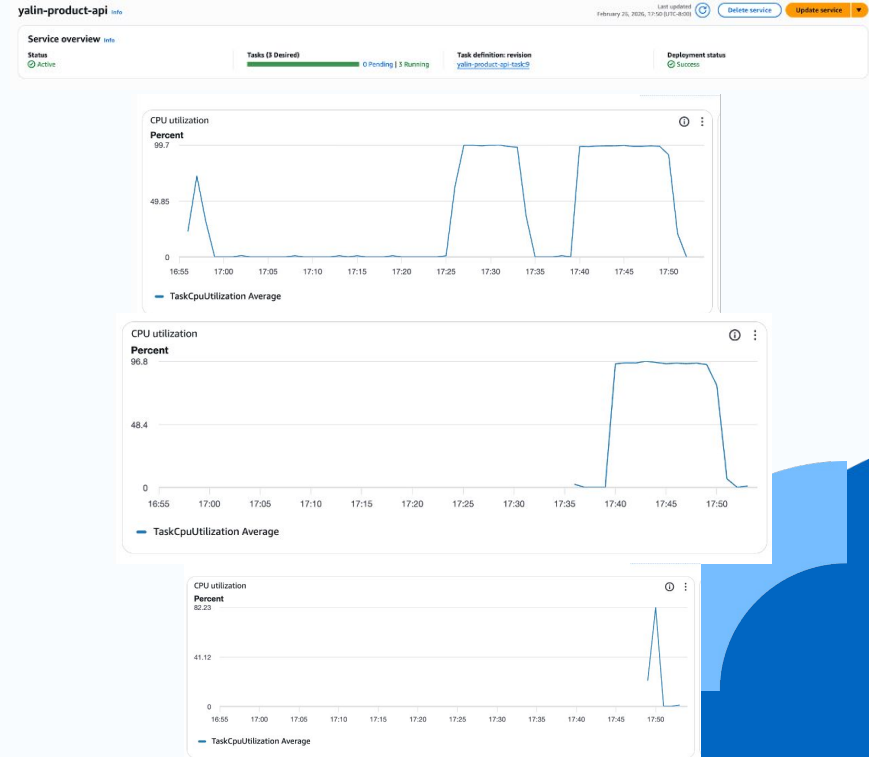


Step I: HW 6 Results

Part III: Horizontal Scaling with Auto Scaling

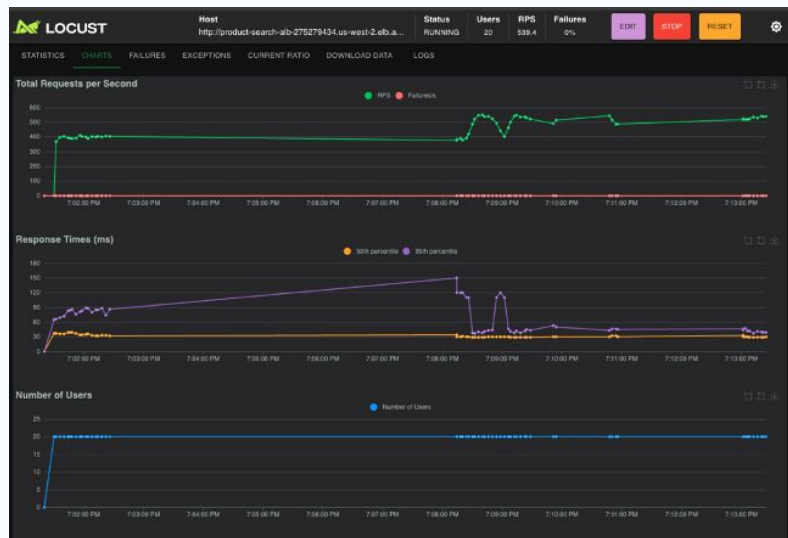
Test 1: Breaking point

- Users: 20
- Spawn Rate: 20 users/second
- Run Time: 12 minutes



Part III: Horizontal Scaling with Auto Scaling

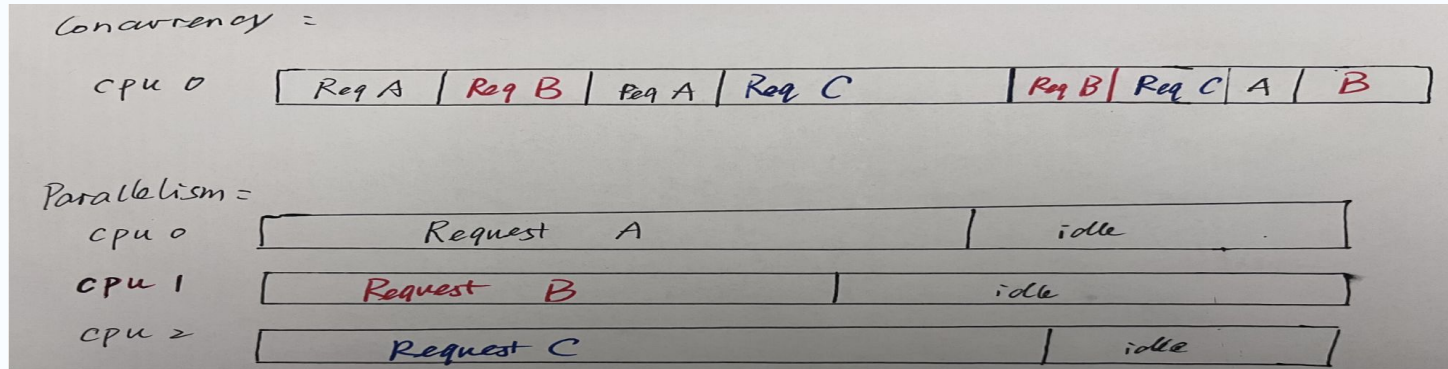
- Manually Stop a Task



Step II: What did you find valuable so far?

1. Concurrency and Distributed Systems

- Definition / HW3 / MapReduce



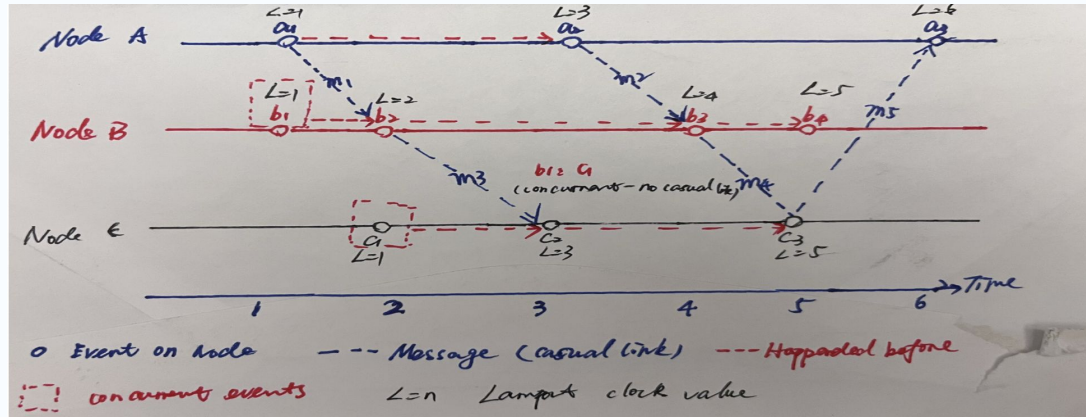
2. Infrastructure as Code and Reproducibility

- Terraform / Docker / ECS, ECR

Step II: What did you find valuable so far?

3. Time, Ordering, and the Illusion of a Global Clock

- Lamport's logical clocks / Race Condition



4. APIs as Contracts Between Systems

- HTTP return codes / REST-style APIs / RPC-style APIs

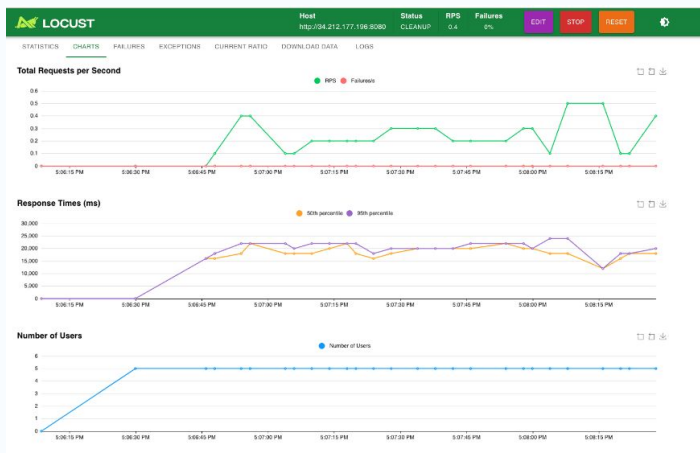
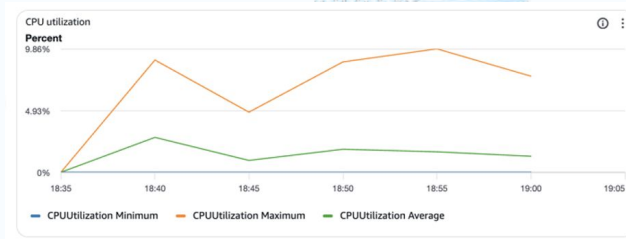
Step III: Crashing and recovering!

1. Problematic Deployment

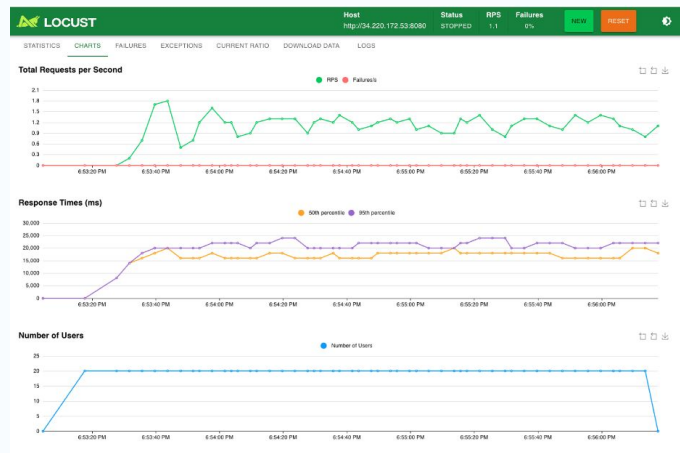
- How the Failure Was Introduced?
- Two load tests to show the problem

Test A (Baseline Load) - User:5 / Spawn rate: 5 / RunTime: 2 mins

Test B (Higher Load) - User:20 / Spawn rate: 10 / RunTime: 3 mins



Test A



Test B

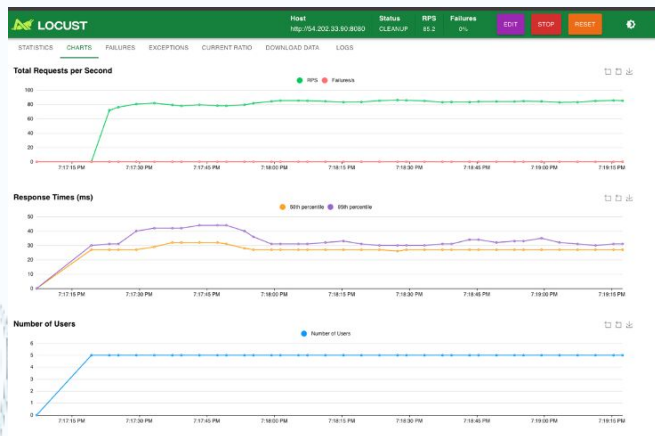
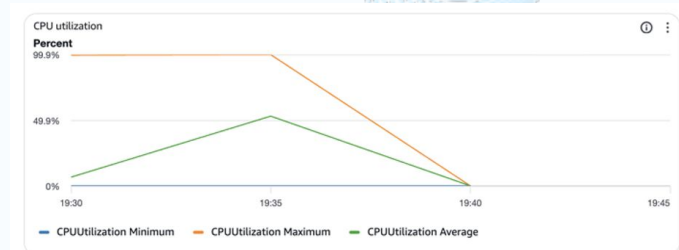
Step III: Crashing and recovering!

2. Fix: How I Improved the System

- Resilience Mechanisms Implemented
Fail Fast / Bulkhead / Circuit Breaker
- Two load tests

Test A (Baseline Load) - User:5 / Spawn rate: 5 / RunTime: 2 mins

Test B (Higher Load) - User:20 / Spawn rate: 10 / RunTime: 3 mins



Test A



Test B